

Octopus

History-Free Gradient Orthogonalization for Continual Learning in MLLMs

Yuehao Liu et al. · CVPR 2026

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Agenda

1. **Problem** — Catastrophic Forgetting in MLLM Continual Learning
2. **Key Insight** — Taylor Expansion → True Lossless Condition
3. **Method** — GPWC & HiFGO
4. **Method** — Two-stage Finetuning
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9. **Ablation** — Order Robustness
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Problem: Catastrophic Forgetting in MLLM CL

Setting

- Sequential tasks $T_1, T_2, \dots, T_N$ arrive one by one
- Previous data **inaccessible** after task ends
- Single LoRA adapted sequentially on MLLM

Prior Work Fails

Method	Problem
Architecture-based (HiDe-LLaVA)	Inference overhead, generalization drop
Rehearsal-based	Privacy risk, memory cost
Regularization-based (O-LoRA)	Param. orthogonality $\neq$ gradient interference prevention
OGD	Requires historical data

Core Challenge

$$\mathcal{L}_{\mathcal{D}_1}(\theta'_1) \ll \mathcal{L}_{\mathcal{D}_1}(\theta'_1 + \theta_2)$$

- New task learning **destroys** old task performance.

Key Insight: What is the True Lossless Condition?

Lossless Condition (Eq. 4)

Task 1 performance must not degrade after learning Task 2:

$$\mathcal{L}_{\mathcal{D}_1}(\theta'_1) = \mathcal{L}_{\mathcal{D}_1}(\theta'_1 + \theta_2)$$

Taylor Expansion (Eq. 5)

$$\mathcal{L}_{\mathcal{D}_1}(\theta'_1 + \theta_2) \approx \mathcal{L}_{\mathcal{D}_1}(\theta'_1) + \left\langle \frac{\partial \mathcal{L}_{\mathcal{D}_1}(\theta'_1)}{\partial \theta'_1}, \theta_2 \right\rangle$$

LoRA weights are small $\rightarrow$ higher-order terms negligible

Result (Eq. 6)

$$\boxed{\left\langle \frac{\partial \mathcal{L}}{\partial \theta'_1}, \theta_2 \right\rangle = 0}$$

Gradient orthogonality is the true lossless condition

Why NOT Parameter Orthogonality?

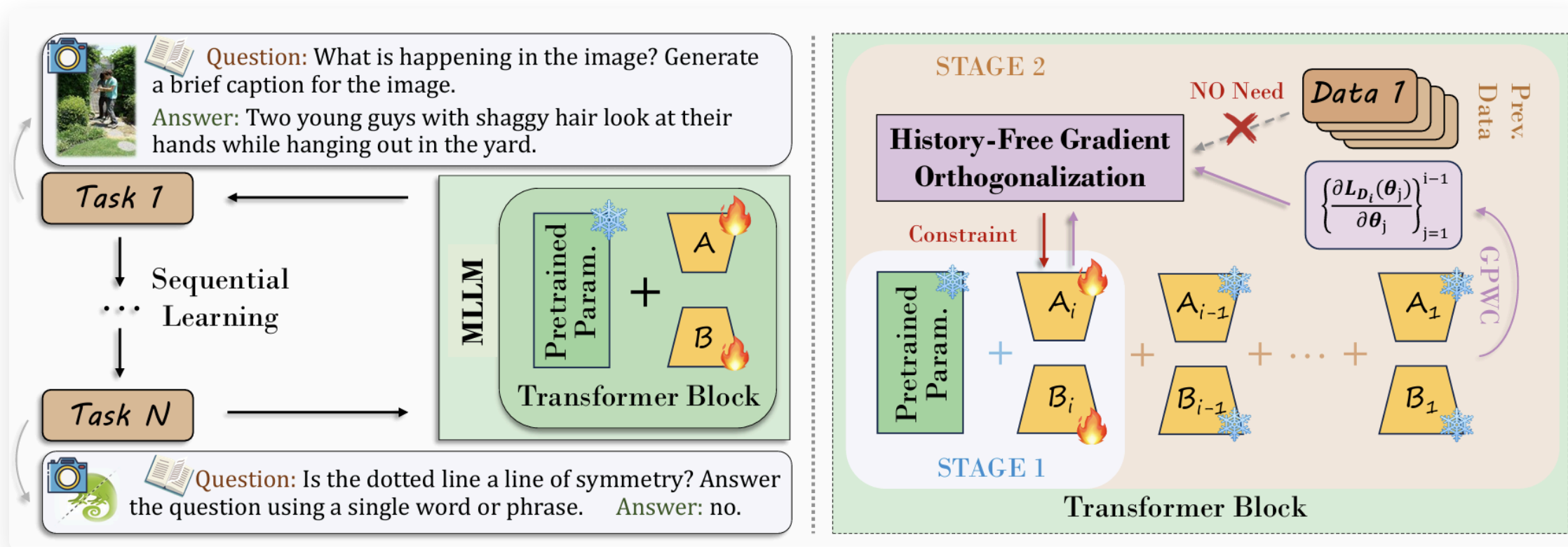
EWC / O-LoRA

- Enforce $\langle \theta_1, \theta_2 \rangle = 0$
- $\theta_1 = \text{accumulated trajectory}$, not gradient direction
 $\rightarrow$ Does **NOT** prevent gradient interference

Why NOT Existing Gradient Orthogonality (OGD)?

- Requires old data $\mathcal{D}_1$ to compute gradients
 $\rightarrow$ Privacy & memory issue
- Converged weights $\rightarrow$ gradient **weak & oscillatory**

Solution: GPWC + HiFGO



Liu et al., "Octopus" · CVPR 2026 · Figure 2: Overview of HiFGO and Two-stage Finetuning

GPWC: Gradients of Previous parameters Within Current data

Key Idea

No access to old data $\mathcal{D}_j$ — instead, pass **current data** through **previous weights**:

$$\text{GPWC} = \frac{\partial \mathcal{L}_{\mathcal{D}_i}(\theta'_j)}{\partial \theta'_j}$$

- $\theta'_j = W_0 + \sum_{m=1}^j \theta_m$: accumulated past weights
- Evaluated on $\mathcal{D}_i \rightarrow$ non-zero, reliable signal

Why does it work?

- Converged θ'_j on **new data** gives a meaningful gradient direction
- Reveals where the update **conflict** with past knowledge

Orthogonality Loss (Eq. 7)

$$\mathcal{L}_{\text{orth}}(\theta_i) = \sum_{j=1}^{i-1} \left(\frac{\partial \mathcal{L}_{\mathcal{D}_i}(\theta'_j)}{\partial \theta'_j} \right)^T \theta_i$$

Minimizing $\mathcal{L}_{\text{orth}} \rightarrow 0$ forces:

$$\theta_i \perp \text{GPWC}_j \quad \forall j < i$$

No historical data needed.

HiFGO = History-Free Gradient Orthogonalization
Use current data + past weights to approximate the true gradient constraint

Two-stage Finetuning: Why Needed?

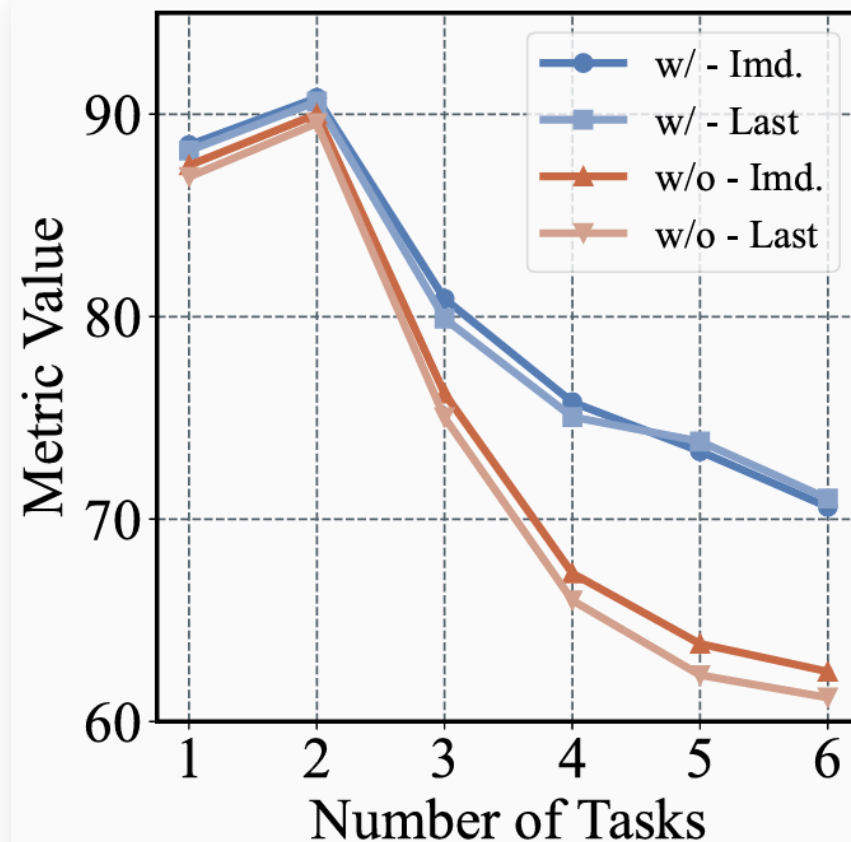
Problem with Naive Regularization

Applying HiFGO + L2 from the **start**:

- Shrinks effective parameter search space
- $\mathcal{L}_{ce}$ vs $\mathcal{L}_{orth}$ **compete** → stuck in suboptimal local minima
- Performance collapses (especially after task 3+)

w/o 2-stage: **61.18%**

w/ 2-stage: **71.01%** (+9.83%p)



(b)

Two-stage Finetuning: Algorithm

Stage 1 — Free Adaptation

Disable all regularization.

Train freely to reach optimal manifold:

$$\mathcal{L}_1 = \frac{1}{|\mathcal{D}_i|} \sum \mathcal{L}_{\text{ce}}(f_{\theta'_{i,1}}(x_k), y_k)$$

- Goal
: Enter the neighborhood of the optimal solution
- No HiFGO, no L2

Stage 2 — Regularized Refinement

Initialize from Stage 1 weights. Activate HiFGO + L2:

$$\mathcal{L}_2 = \frac{1}{|\mathcal{D}_i|} \sum \left(\mathcal{L}_{\text{ce}} + \lambda_1 \mathcal{L}_{\text{orth}} + \lambda_2 \mathcal{L}_{\text{norm}} \right)$$

- $\mathcal{L}_{\text{orth}}$: gradient orthogonality (HiFGO)
- $\mathcal{L}_{\text{norm}}$: L2 regularization to keep $\|\theta_i\|$ small (validates Taylor approximation)
- Fine-tune near the optimum **without** competition between objectives

Experimental Setup: UCIT Benchmark

6 Sequential Domains

#	Dataset	Domain
1	ImageNet-R	Image Classification
2	ArXivQA	Academic QA
3	VizWiz	Visual QA
4	IconQA	Icon Recognition QA
5	CLEVR-Math	Math Reasoning
6	Flickr30k	Image Captioning

Backbone: LLaVA (MLLM) + Single LoRA

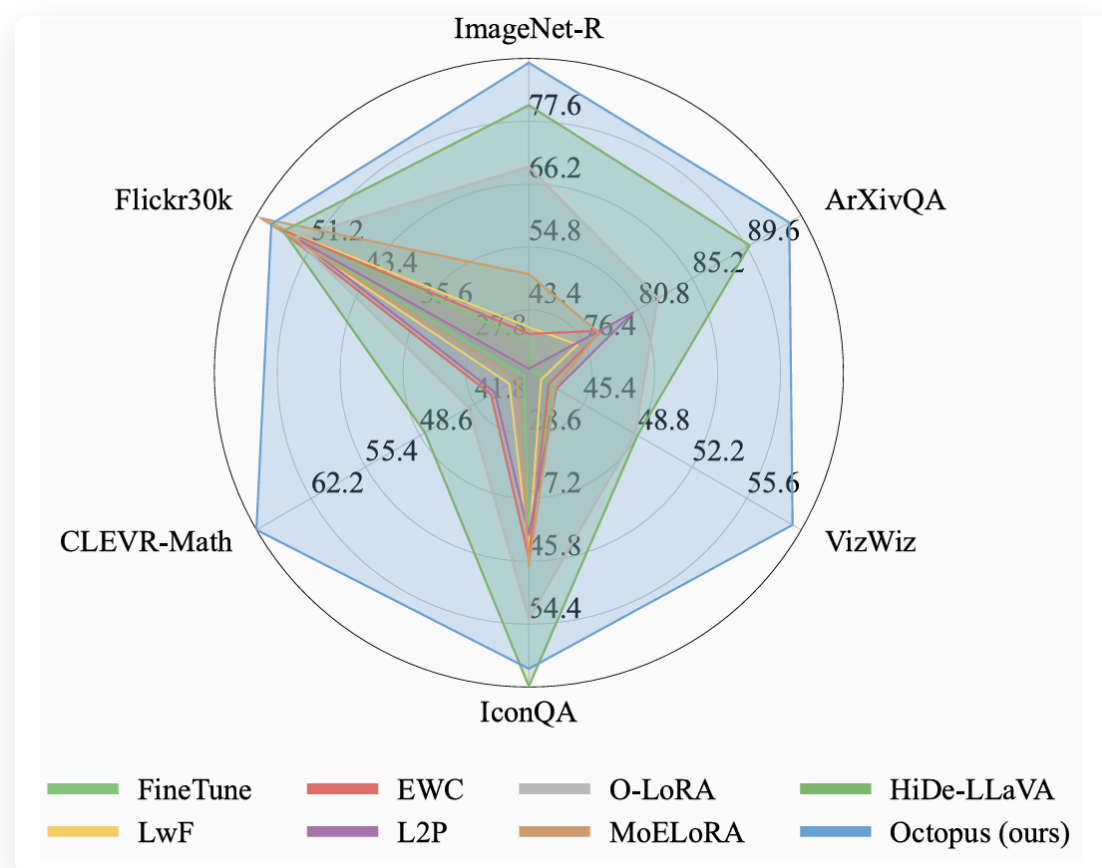
Evaluation Metrics

Metric	Description
Last	Avg. accuracy after all 6 tasks (<i>primary</i>)
Avg	Avg. accuracy across all stages
Imd.	Accuracy right after learning each task (upper bound)
BWT	Backward Transfer — positive = knowledge gain

Baselines

- Architecture-based: HiDe-LLaVA, MoILE
- Rehearsal-based: Vanilla Rehearsal
- Regularization-based: EWC, O-LoRA, LwF
- Gradient-based: OGD, InfLoRA

Main Results: Radar Chart



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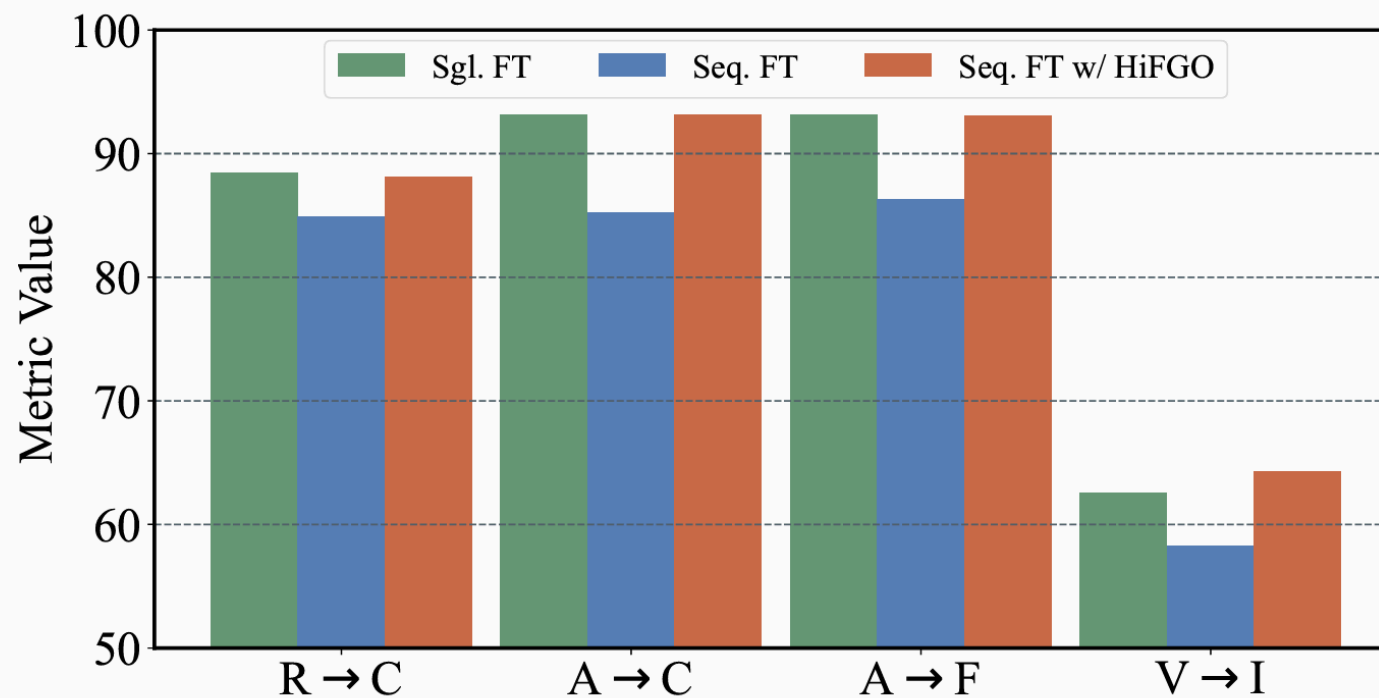
Main Results: Quantitative Comparison

	Method	Replay	ImageNet-R	ArXivQA	VizWiz	IconQA	CLEVR-Math	Flickr30k	Average
	Zero-shot	-	17.20	52.16	41.07	18.70	18.63	46.99	32.46
	Multi-task	-	89.53	92.20	62.70	61.86	70.40	58.49	72.53
Avg	Sequential Finetune	-	49.31	78.40	50.48	53.44	55.53	57.95	57.52
	Vanilla Rehearsal	✓	88.71	86.13	54.15	61.76	70.53	58.13	69.90
	▽ Architecture-based								
	L2P [50]	✗	41.52	82.32	51.98	52.21	43.16	52.77	53.99
	MoELoRA [7]	✗	64.49	82.42	49.54	56.87	56.35	58.34	61.33
	HiDe-LLaVA [17]	✗	85.70	92.70	54.10	66.87	59.12	55.15	68.94
	▽ Regularization-based								
	LwF [27]	✗	55.60	79.86	53.23	54.87	56.51	56.34	59.40
	EWC [24]	✗	54.23	80.13	53.14	55.06	57.52	55.94	59.34
	O-LoRA [48]	✗	75.26	86.73	55.86	58.47	57.38	53.52	64.54
	Octopus (ours)†	✗	89.69	91.34	<u>61.04</u>	<u>60.18</u>	68.09	<u>57.61</u>	71.33
	Octopus (ours)	✗	<u>88.41</u>	93.30	61.65	59.91	66.32	56.87	<u>71.08</u>
Last	Sequential Finetune	-	37.63	72.33	43.47	41.70	35.63	57.95	48.12
	Vanilla Rehearsal	✓	88.43	85.36	51.50	60.10	67.13	58.13	68.44
	▽ Architecture-based								
	L2P [50]	✗	32.73	80.41	43.72	42.16	39.25	52.77	48.51
	MoELoRA [7]	✗	49.87	77.63	43.65	46.40	36.47	58.34	52.06
	HiDe-LLaVA [17]	✗	80.50	<u>89.83</u>	48.78	62.90	47.97	55.15	64.19
	▽ Regularization-based								
	LwF [27]	✗	40.27	75.93	42.76	44.38	37.43	56.34	49.52
	EWC [24]	✗	39.05	77.88	43.24	45.33	39.72	55.94	50.20
	O-LoRA [48]	✗	69.36	82.42	48.64	53.66	42.53	53.52	58.36
	Octopus (ours)†	✗	88.80	89.83	56.44	59.90	70.60	<u>57.11</u>	<u>70.45</u>
	Octopus (ours)	✗	<u>88.20</u>	93.03	58.46	<u>60.50</u>	<u>69.00</u>	56.87	71.01

Liu et al., "Octopus" · CVPR 2026 · Table 1: UCIT Benchmark Results

	vs. SOTA (HiDe-LLaVA)	vs. Rehearsal	vs. O-LoRA
Octopus	Avg +2.14%p / Last +6.82%p	Last 71.01% vs 68.44% (no data)	Avg +6.54%p / Last +12.65%p

Ablation: HiFGO Recovery Capability



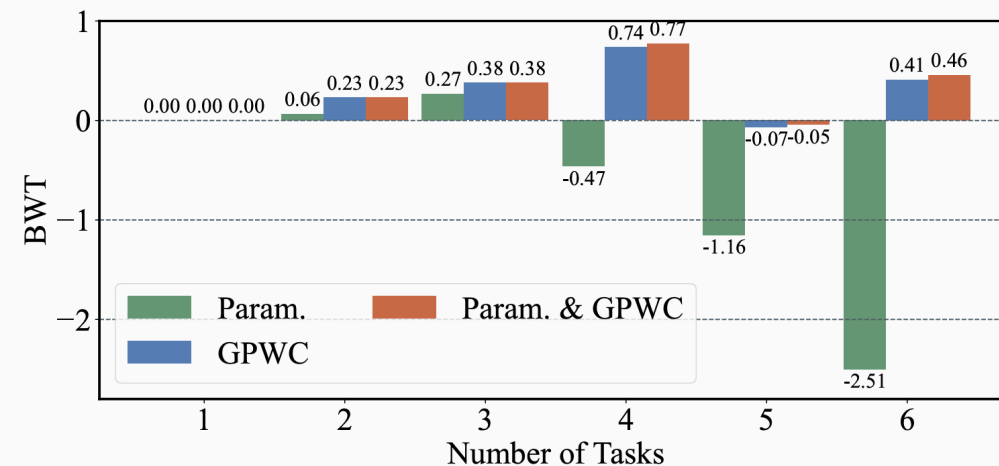
Liu et al., "Octopus" · CVPR 2026 · Figure 3: Task 1 performance after learning Task 2

Seq. FT (no reg.) → Task 1 collapses | Seq. FT w/ HiFGO → recovers to Sgl. FT level (single-task upper bound)

Ablation: What to Orthogonalize?

Settings		ImageNet-R	ArXivQA	VizWiz	IconQA	CLEVR-Math	Flickr30k	Average	BWT
∇ Constrain current parameters orthogonal to <i>what</i>									
Prev. Param.	Imd.	88.47	91.73	60.74	56.73	61.33	56.29	69.22	
	Last	86.67	92.27	50.62	57.00	57.40	56.29	66.71	-2.51
GPWC	Imd.	88.47	93.13	61.04	60.47	63.63	56.87	70.60	
	Last	88.20	93.03	58.46	60.50	69.00	56.87	71.01	+0.41
Prev. Param. & GPWC	Imd.	88.47	93.20	60.50	60.77	63.60	56.99	70.59	
	Last	88.27	92.90	59.01	60.27	68.83	56.99	71.04	+0.45
∇ Compute Gradients on <i>what</i>									
Curr. Task (GPWC)	Imd.	88.47	93.13	61.04	60.47	63.63	56.87	70.60	
	Last	88.20	93.03	58.46	60.50	69.00	56.87	71.01	+0.41
Prev. Task	Imd.	88.47	93.53	61.53	61.67	65.70	56.31	71.20	
	Last	86.50	73.43	51.47	58.57	46.40	57.15	62.25	-8.95

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Orthogonalize target	Last	BWT
Param. vectors ($\theta_1 \perp \theta_2$)	66.71%	-2.51
GPWC — current data (ours)	71.01%	+0.41
Prev. task gradient (old data, OGD-style)	62.25%	-8.95

Ablation: 2-stage & λ Sensitivity

Method		ImageNet-R	ArXivQA	VizWiz	IconQA	CLEVR-Math	Flickr30k	Average	BWT
w/ two-stage finetuning	Imd.	88.47	93.13	61.04	60.47	63.63	56.87	70.60	+0.41
	Last	88.20	93.03	58.46	60.50	69.00	56.87	71.01	
w/o two-stage finetuning	Imd.	87.47	92.53	48.61	40.70	49.90	55.62	62.47	-1.29
	Last	86.90	92.20	45.96	38.90	47.53	55.62	61.18	

Liu et al., "Octopus" · CVPR 2026 · Table 3: Two-stage ablation

w/ 2-stage: 71.01% | w/o: 61.18%

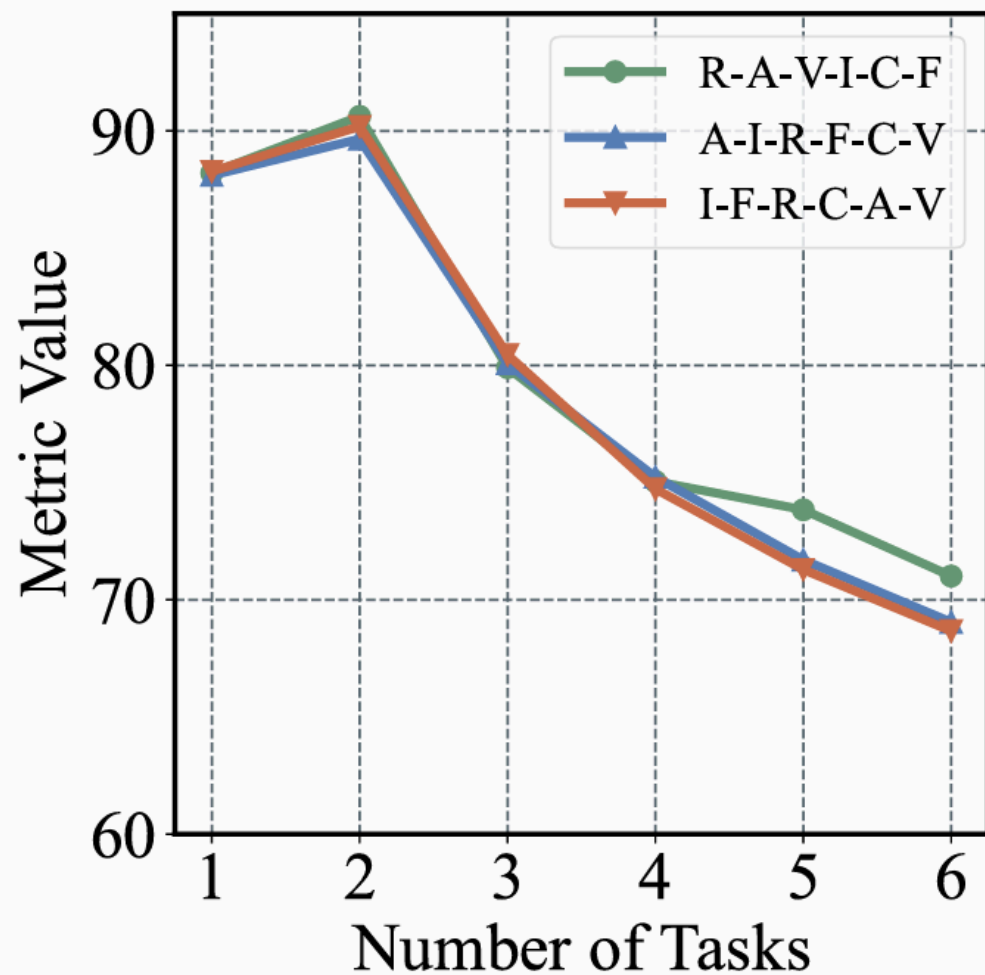
Method		ImageNet-R	ArXivQA	VizWiz	IconQA	CLEVR-Math	Flickr30k	Average	BWT
w/o norm-regularization	-	Imd.	89.03	93.37	63.31	65.03	73.07	56.33	73.36
		Last	81.50	81.43	47.65	57.57	63.27	57.25	64.77
w/ norm-regularization	$\lambda = 2e-3$	Imd.	88.73	92.03	61.90	62.30	70.30	56.78	72.01
		Last	85.07	86.47	54.93	61.63	67.07	56.78	68.65
	$\lambda = 5e-3$	Imd.	88.23	93.87	62.34	63.37	68.27	57.13	72.20
		Last	87.40	91.23	58.20	62.13	68.23	57.13	70.72
	$\lambda = 1e-2$	Imd.	88.47	93.13	61.04	60.47	63.63	56.87	70.60
		Last	88.20	93.03	58.46	60.50	69.00	56.87	71.01
	$\lambda = 3e-2$	Imd.	87.93	89.60	54.92	43.67	51.57	57.01	64.12
		Last	88.07	92.93	49.46	54.57	60.17	57.01	67.03

Table 4. Effectiveness of norm regularization. We report performance under both w/ and w/o settings in terms of *Imd.* and *Last*.

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- Too small $\rightarrow$ high forgetting (BWT -8.59)
- Too large $\rightarrow$ low plasticity (Imd. 64.12%)
- Optimal: $\lambda = 1e-2$

Ablation: Order Robustness



(a)

Order	R-A-V-I-C-F	A-I-R-F-C-V	I-F-R-C-A-V
Last	71.01	69.04	69.16
Avg	71.08	70.04	70.00

Liu et al., "Octopus" · CVPR 2026 · Table 5: Order robustness results

Conclusion

Contributions

1. **Theoretically proved** (Taylor expansion) that gradient orthogonality is the true lossless condition for CL
2. **GPWC** — computes meaningful gradient signals from previous weights using *current data*, removing dependence on historical data
3. **2-stage finetuning** — separates free learning from constrained refinement, eliminating regularization-plasticity competition

Pros & Cons

Pros

- No historical data (privacy & memory free)
- Single LoRA — no extra modules
- Order-invariant robustness

Cons

- GPWC cost grows linearly with task count
- λ hyperparameter sensitivity
- Limited CL scenario coverage

Thank You

Questions?